Thomas Niedermayer

Data Engineer, Iknaio Cryptoasset Analytics GmbH

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Data engineer and data scientist with a background in mathematics, working with blockchain data since 2020. At Iknaio, works on blockchain data in general, from the pipeline that brings raw data into GraphSense, the open-source platform behind Iknaio's investigation products (100B+ rows), to automated tracing of funds. First author of a peer-reviewed paper on detecting financial bots on Ethereum. Speaker and trainer for law enforcement.

Key Achievements

- **Blockchain data.** Builds and runs ingestion and transformation from blockchain nodes into Iknaio's own cluster, 100B+ rows across Bitcoin, Ethereum and other ledgers. Works on automated tracing of funds.
- **Research.** First author of a peer-reviewed paper on financial bot detection on Ethereum at The Web Conference (WWW '24), using clustering, classification and explainable AI. Co-author of a 2026 study on how reliable Bitcoin address clustering is in law enforcement cases.
- **Work with public authorities.** Speaker at the INTERPOL New Technologies Forum 2026 (Operation Alice, joint session with CFLW) and at a CEPOL workshop on terrorist financing (Athens, 2026). Trainer in Iknaio courses for law enforcement and financial investigators.

Professional Experience

2023 – Present **Data Engineer**, *Iknaio Cryptoasset Analytics GmbH, Vienna*. Blockchain data in general, from data engineering of GraphSense to automated tracing of funds for investigations.

2022 – 2023 **Master's Thesis Student**, *Complexity Science Hub, Vienna*. Thesis on detecting financial bots on Ethereum with graph algorithms and clustering, published at WWW '24. ETL pipelines and data modelling (100M+ rows).

2020 – 2022 **Blockchain Bot Developer**, *self-employed*. Developed and operated automated trading bots on the Harmony blockchain.

2015 – 2020 **Data Analyst and Internships**, *Erste Asset Management, Deutsche Bank, T-Mobile Austria, Erste Bank, Gradient Zero*. Data pipelines and dashboards in R, risk simulation for investment funds, deep learning and privacy-preserving synthetic data.

Education

2023 **MSc in Data Science**, *TU Wien*.

2020 **BSc in Mathematics**, *University of Vienna*. Focus on optimisation and mathematical modelling.

Publications

1. Müller, L., Elsner, J., Niedermayer, T., Haslhofer, B., Goger, T., Kühl, N., & Rückert, C. (2026). How reliable is the multi-input heuristic for Bitcoin address clustering in law enforcement contexts? *arXiv preprint*. arXiv:2607.07414

2. Niedermayer, T., Saggese, P., & Haslhofer, B. (2024). Detecting financial bots on the Ethereum blockchain. In *Companion Proceedings of the ACM Web Conference 2024 (WWW '24)*. arXiv:2403.19530

Selected Projects

2026	**graphsense-clustering**, *open source*. The address clustering engine of GraphSense as a stand-alone Rust package. Clusters all 1.4 billion Bitcoin transactions on a laptop.
2025	**Data pipeline for CoW DAO**, *grant by CoW DAO*. Weekly ingest of solver metrics from node RPCs into Dune Analytics, with a public dashboard.

Skills

Languages	German (native), English (C1), Chinese (B1).
Programming	Python, SQL, Rust, Scala, R.
Data stack	Spark, Delta Lake, DuckDB, Cassandra, PostgreSQL, Dagster, dbt, Docker, Dune Analytics.
ML	PyTorch, scikit-learn, graph algorithms, explainable AI.
Certification	dbt Analytics Engineering.